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# FINANCEGPT: PRECISION FINANCIAL FORECASTING AND BUDGETING FOR SMARTER INVESTMENT STRATEGIES

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**ABSTRACT:**

FinanceGPT is an AI-powered financial management tool designed to streamline expense tracking and assist users in making informed decisions on loans and term insurance. The system comprises two primary modules: Expense Tracker and Loan & Insurance Predictor.The Expense Tracker module processes user-submitted bills and receipts, employing Optical Character Recognition (OCR) to extract textual data. In cases where the original text is unavailable, alternative techniques retrieve relevant financial information. A BERT-based information retrieval model categorizes and groups expenses into predefined categories such as entertainment, utilities, and meals, enabling users to analyze their spending patterns effectively. Additionally, a threshold-based alert system notifies users when predefined spending limits are exceeded, helping them maintain financial discipline. All transaction data is stored in a database for long-term tracking and trend analysis.

The Loan & Insurance Predictor module evaluates users' financial profiles to assess loan eligibility and recommend suitable term insurance coverage. By analyzing salary details, existing liabilities, and other financial credentials, the system computes the Fixed Obligations to Income Ratio (FOIR) to determine the user's borrowing capacity and affordable EMI range. This information aids users in making informed borrowing decisions. Furthermore, the module generates personalized insurance recommendations based on the user's financial liabilities, ensuring adequate coverage.By integrating AI-driven financial analytics, FinanceGPT enhances budget management, optimizes loan decisions, and provides tailored insurance insights, offering a comprehensive solution for personal finance management.

Keywords: Finance, OCR, BERT, loan, insurance.

## I INTRODUCTION

Sustainability over the long-term and stability for security purposes mandate good finance handling. Only a select group of people make it to capture the time and expertise required in handling money efficiently, yet all planning should help handle the budget, borrowing, and policies about insurance. IT enterprises as well as firms that are finances-oriented designed sophisticated systems where machine intervention nullified the cost audit, budgets making, as well as making major decisions.Artificial intelligence (AI) and machine learning (ML) technology have made way for autonomous financial management systems. Technologies such as Optical Character Recognition (OCR) for extracting text [1], Natural Language Processing (NLP) for analyzing context, and deep learning architectures such as BERT for classifying text [2] have significantly enhanced financial analytics. These technologies allow for the creation of intelligent systems that offer real-time insights, automated expense tracking, and customized financial recommendations.

The paper introduces FinanceGPT, a holistic AI-based financial management model incorporating expense tracking, loan evaluation, and term insurance prediction. The system reads financial information from bills and receipts through OCR, classifies expenses through BERT-based NLP, and offers interactive data visualization to enable users to examine spending patterns. It also assesses loan eligibility by calculating the Fixed Obligations to Income Ratio (FOIR) and deriving the maximumpossible EMI based on users' financial information [3]. The model also provides term insurance suggestions that are tailored to users' age, income, and financial obligations [4]. By

harnessing AI-based automation, FinanceGPT enables users to make well-informed financial decisions, maximize spending efficiencies, and achieve greater financial stability.
The model under consideration acts as an intelligent personal financial advisor, providing a hassle-free and data-centric way of managing budgets, loan evaluation, and insurance planning. The model, in the end, brings individuals to be masters of their money, hence making the right decisions and easy and convenient steps towards achieving financial stability [9].

## II LITERATURE REVIEW

Based on their results, RF models performed better than conventional credit scoring techniques in loan eligibility prediction. These improvements can make financial planning chatbots more proficient at lending recommendations.With user-specific information like age, risk tolerance, and financial goals, Ghosh et al. (2023) [16] explored AI-driven investing and insurance recommendation methods.
Nonetheless, integrating such models in chatbot applications was not easy, particularly in terms of explainability of the model and user trust in financial applications [12]. Natural Language Processing in Financial Systems, Natural Language Processing (NLP) deployment is vital in financial systems dealing with unstructured text data. Devlin et al.'s Bidirectional Encoder Representations from Transformers (BERT) model significantly improves contextual understanding in financial transactions. Lee and Yang (2020) [13], [14] applied BERT for classifying money transactions into highly accurate spending groups even with less training data. While BERT enhances cost management, it presents high computation loads that can diminish real-time execution. Model compression techniques such as distillation help overcome these disadvantages and enhance BERT-based finance applications for their efficiency.
For Loan Eligibility Prediction,Bhatnagar and Rai (2022) [15] studied ensemble models like Random Forest for predicting loan eligibility based on financial attributes like credit scores, income, debt-to-income ratio, and employment. Their results indicated that RF models were more effective than conventional credit scoring models in predicting loan eligibility. These improvements can assist financial planning chatbots in enhancing their loan suggestion.
For AI-Driven Investment and Insurance Advisory Ghosh et al. (2023) [16] experimented with AI-driven investment strategies based on user-specific information like age, risk tolerance, and investment horizon.Their research demonstrated how models of machine learning would enable personalized investment advice, resulting in higher customer satisfaction and return on investment.
For insurance suggestion, Jain and Rao (2021) [17] introduced a model that individualizes term insurance policies according to user characteristics like income, age, dependents, and medical history. They emphasized that transparency and interpretability of the model are critical in order to instill confidence in AI-driven financial decisions [18].
For Next-generation Financial Chatbots, New NLP models like LLaMA are proposed to improve the efficiency of chatbots in next-generation financial conversation. Future studies involve incorporating multi-modal financial chatbots that deal with text, voice, and visual inputs for better user experience. Hybrid methods involving machine learning (e.g., Random Forest) and deep learning methods could also result in improved and personalized financial recommendations [19]. Increasing the flexibility of chatbots and combining machine learning algorithms like Random Forest with deep learning approaches would result in more precise, tailored, and useful financial advice.
For AI Detection of Financial Fraud with financial services increasingly going digital, fraud detection is a newly emerging field of research. Xu et al. (2022) investigated the application of deep learning methods like Graph Neural Networks (GNNs) to detect fraudulent transactions in real-time. Their study revealed that GNNs are good at detecting concealed patterns among fraudsters with great accuracy, even superior to conventional rule-based fraud detection systems. Apart from that, AI hybrid models integrating Random Forest and deep learning have been put forward to support fraud detection with improved accuracy while being computationally lightweight [20]. Among the challenges, it is one of the prominent issues to be able to have high model complexity along with interpretability as financial institutions require transparent decision-making.
Sentiment Analysis for Financial Decisions of recent research has shown the effectiveness of sentiment analysis in financial prediction. Kim et al. (2021) used transformer-based models to study financial news and social media postings, forecasting stock market trends more accurately. Sentiment analysis has also been implemented in financial chatbots to optimize loan and investment suggestions by measuring market mood. While these methods are promising, difficulty exists in dealing with misinformation and noisy data sources and thus require robust filtering methods for the delivery of credible insights. Ethical and Regulatory Challenges in AI-Driven Finance in spite of the progress of AI-based financial systems, regulatory issues and ethical issues remain ongoing. GDPR and other data protection regulations have strict demands on financial AI tools, particularly user data security and avoiding bias. Brown et al. (2023) highlighted the importance of explainable AI (XAI) to promote fairness and explainability in automated financial decisions. Bias in the training data will produce discriminatory lending approval or misserving risk determinations, the implications of which will require a consistent research interest in bias-correcting algorithms and equitable AI models. Financial AI systems of the future need to hold ethical principles over efficiency and precision.

## III PROPOSED SYSTEM

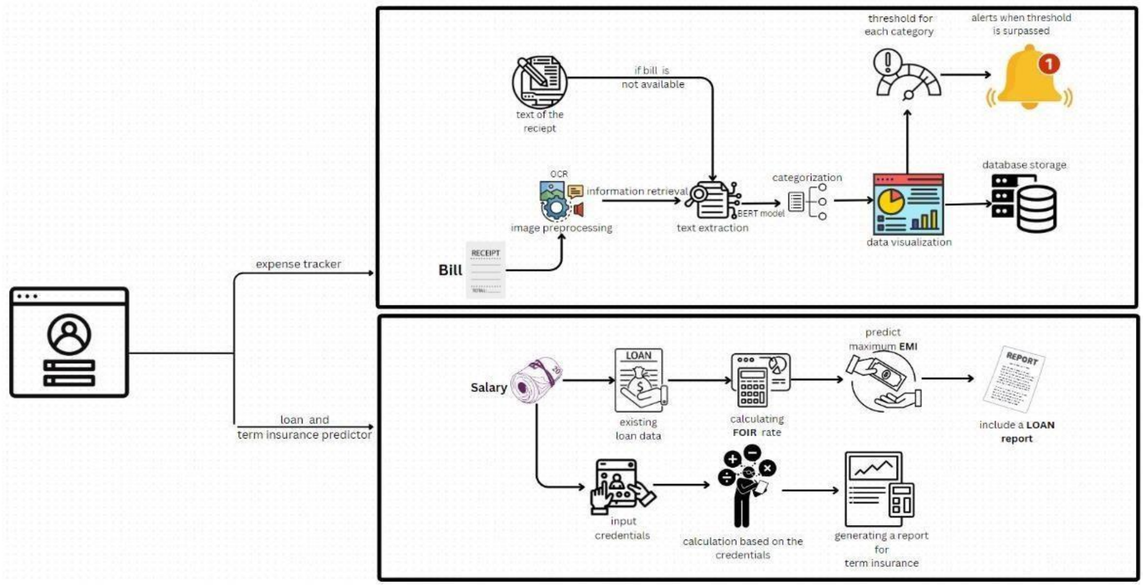



**Figure 1**. Proposed Architecture

The proposed model is a dual-function system, encompassing two primary modules: The Expense Tracker and the Loan & Insurance Predictor. The Expense Tracker applies OCR to scan and process bills and receipts, converting the physical or digital documents into structured data that can be analyzed. The system will be able to categorize expenses correctly and classify them into categories such as groceries, utilities, and leisure by using a BERT-based NLP model. The will allow users to have clear insights into their spending habits, making it easier for them to identify areas where they are overspending. The model also allows users to set spending thresholds for each category, triggering alerts whenever these limits are exceeded. The proactive approach helps in better budgeting and ensures users adhere to their allocated limits. The module further visualizes data through an intuitive interface, providing the user with a lucid and organized view of their expenditure timeline. Loan & Insurance Predictor is further used to assist users in making more informed financial decisions on loans and insurance. It calculates their fixed obligation-to income ratio, taking into account the user's salary, existing financial obligations, and all other credentials relevant in The regard. The ratio is a significant determinant of the user's ability to borrow more loans. In the light of the FOIR, the model can predict the maximum equated monthly installment that the user can reasonably afford.

That way, it gives them realistic insights into their borrowing potential. Another interesting thing with The module is that it designs term insurance according to individual financial profiles and other debts available. The helps individuals wanting their financial future without jeopardizing their current resources. Taken together, the modules work through a holistic model towards proper personal financial management; address their immediate budgetary concerns; and, finally, outline long-term planning issues of finance. Through the automation of data extraction, processing, and analysis, it eliminates the requirement for manual input in the system. The provides wide access and convenience to any user background. The integrated technologies include OCR and BERT that improve the accuracy and efficiency of the system, distinguishing it from the traditional financial management tools. The model empowers the user with his or her own financial decision-making and provides the tools, or the ability to make strategic decisions that can lead him or her toward improve the stability of finance. .

## IV METHODOLOGY

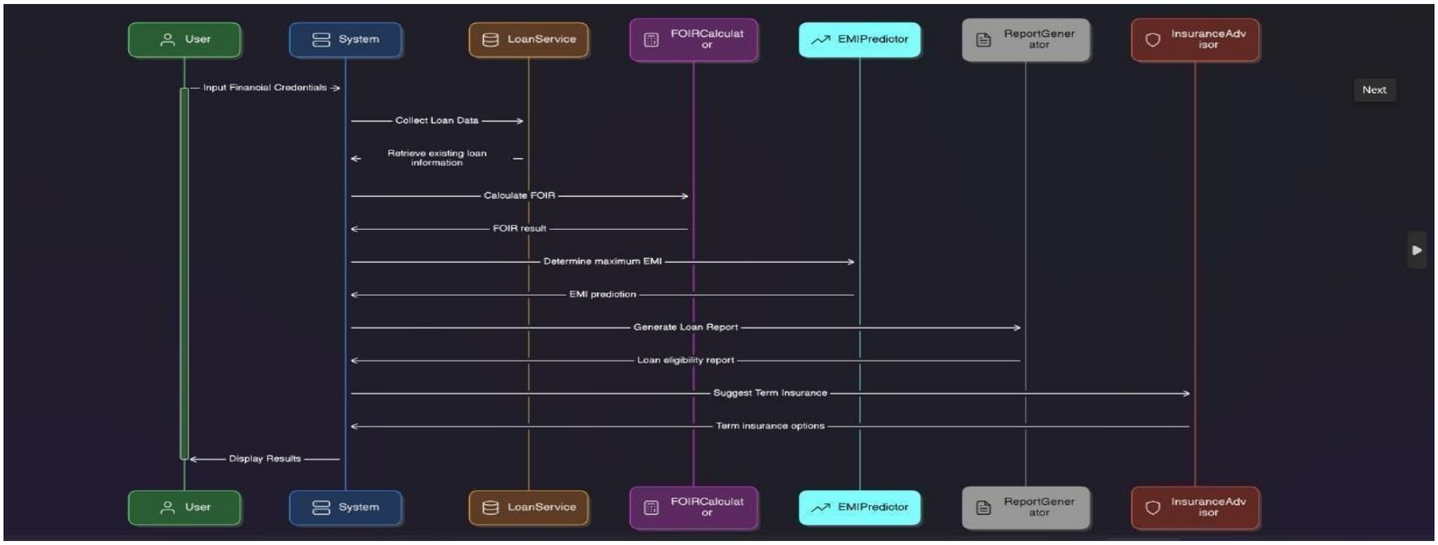



**Figure 2.** methodology diagram for the module

The methodology Figure 2 would be modular and structured in design, to determine the loan eligibility, calculate EMI, and give insurance suggestions. In the user input module, where the financial details like income, liabilities, and existing loan details are input by the user, the process begins. Respective modules then record this information to be processed. The Loan Service Module provides correct calculation of the outstanding liabilities by retrieving and consolidating the user's current loan information from financial accounts. The FOIR Calculator then uses this data to determine the FOIR, a critical indicator of the ability of a borrower to borrow. At The stage, the module EMI Predictor calculates the maximum EMI that can be tolerated by the user and then predicts the EMI for the recommended loan. The Report Generator Module provides a holistic view of the user's borrowing capacity by consolidating the calculated financial parameters into a loan eligibility report. As a drive to provide sufficient financial protection, the Insurance Advisor Module ultimately concludes the user's financial portfolio and suggests suitable term insurance schemes. Finally, the Insurance Advisor module generates term insurance schemes based on the user's financial profile.

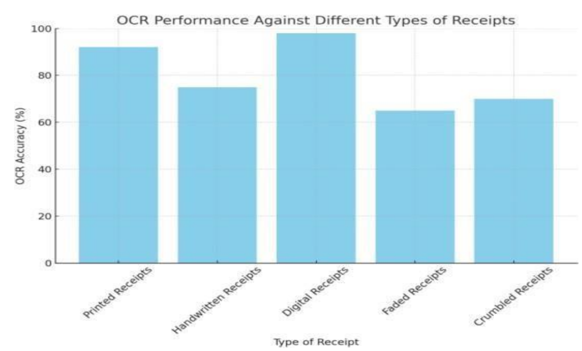



**Figure 4.** OCR Performance of Different Types of Receipts

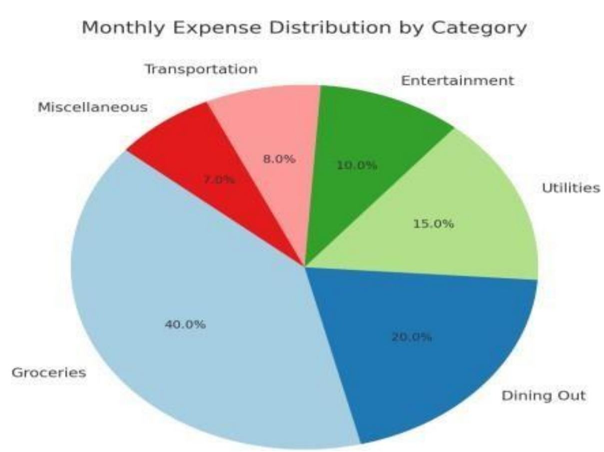



**Figure 5.** Pie Chart of Monthly Expense Distribution by Category

## VI RESULTS AND INFERENCE

The results indicate that it is quite effective in terms of tracking and managing one's expenses by using the expenditure management system. With higher accuracy on the OCR system integrated with the proper categorizing, an automated way for tracking expense is quite trustworthy. The data visualization highly influences user behavior as The visual summarization of spending's promotes financial awareness, and its limitations such as OCR accuracy with low-quality image and also unusual receipt forms are good areas to explore in future.

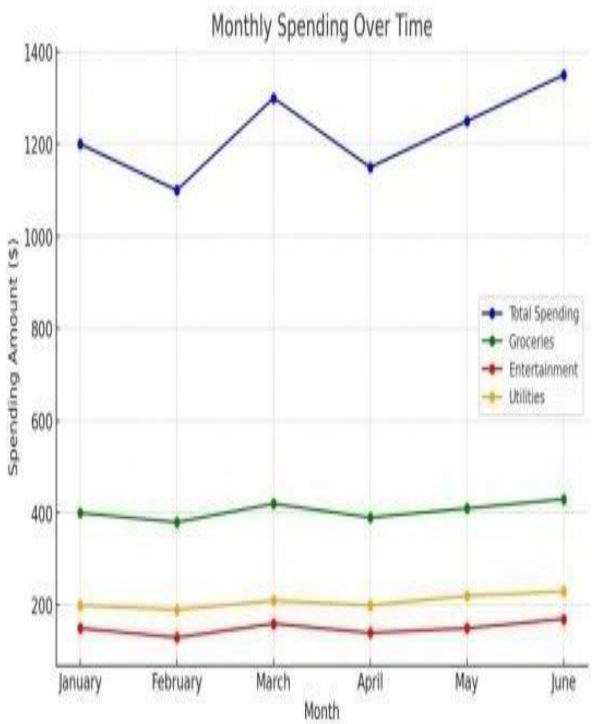
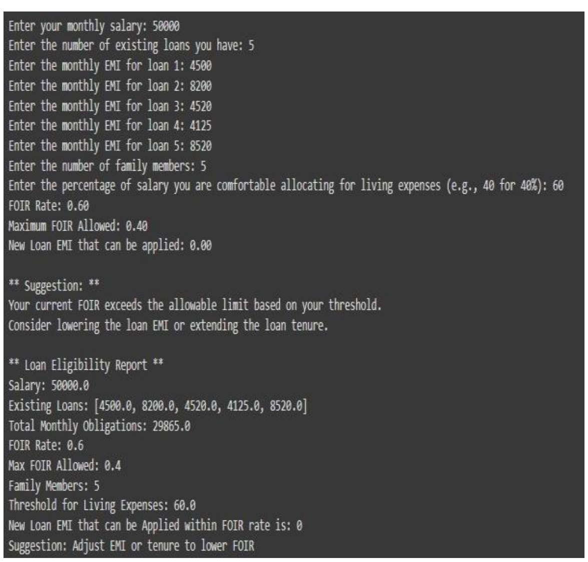


Figure 6: Line Graph of Monthly Spending over Time



```
Enter your monthly salary: 50000
Enter the number of existing loans you have: 5
Enter the monthly EMI for loan 1: 4500
Enter the monthly EMI for loan 2: 8200
Enter the monthly EMI for loan 3: 4520
Enter the monthly EMI for loan 4: 4125
Enter the monthly EMI for loan 5: 8520
Enter the number of family members: 5
Enter the percentage of salary you are comfortable allocating for living expenses (e.g., 40 for 40%): 60
FOIR Rate: 0.60
Maximum FOIR Allowed: 0.40
New Loan EMI that can be applied: 0.00

** Suggestion: **
Your current FOIR exceeds the allowable limit based on your threshold.
Consider lowering the loan EMI or extending the loan tenure.

** Loan Eligibility Report **
Salary: 50000.0
Existing Loans: [4500.0, 8200.0, 4520.0, 4125.0, 8520.0]
Total Monthly Obligations: 29865.0
FOIR Rate: 0.6
Max FOIR Allowed: 0.4
Family Members: 5
Threshold for Living Expenses: 60.0
New Loan EMI that can be Applied within FOIR rate is: 0
Suggestion: Adjust EMI or tenure to lower FOIR
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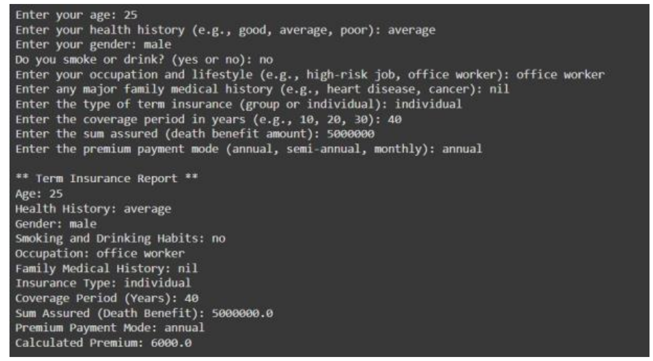
Figure 7: Loan Report Recommendation



```
Enter your age: 25
Enter your health history (e.g., good, average, poor): average
Enter your gender: male
Do you smoke or drink? (yes or no): no
Enter your occupation and lifestyle (e.g., high-risk job, office worker): office worker
Enter any major family medical history (e.g., heart disease, cancer): nil
Enter the type of term insurance (group or individual): individual
Enter the coverage period in years (e.g., 10, 20, 30): 40
Enter the sum assured (death benefit amount): 5000000
Enter the premium payment mode (annual, semi-annual, monthly): annual

** Term Insurance Report **
Age: 25
Health History: average
Gender: male
Smoking and Drinking Habits: no
Occupation: office worker
Family Medical History: nil
Insurance Type: individual
Coverage Period (Years): 40
Sum Assured (Death Benefit): 5000000.0
Premium Payment Mode: annual
Calculated Premium: 6000.0
```

Figure 8: Term Insurance Report Recommendation

## VI CONCLUSION

The dual-function financial management system, therefore, would take all these important needs into an all-embracing account, while aiming at making personal finance management and prudent loan and insurance decision-making as simple as possible for the individual. In The regard, it enables the user to have efficient expense tracking with a predictive loan and insurance model, thereby providing the possibility of personal finances planning and risk alert management. Leverage BERT-based information retrieval and the power of predictive analytics; empowering user capabilities to improve management and control of expenses, further predict loan eligibility, or determine insurance needs. In general, the project seeks the integration of leading methodologies from which to address a great social need: intelligent, data-driven decision support tools for personal and family financial decision making.

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